

A Spatiotemporal Model for Radar HRRP Sequence Recognition

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Abstract—A temporal factor analysis (TFA) model which can describe both the spatial structure and temporal dependency of high range resolution profiles (HRRP) has been proposed for radar HRRP sequence recognition. Variational Bayes method and Bayesian classifier are adopted for model learning and classification task, respectively. The favorable results based on measured data show the effectiveness of this model.

I. INTRODUCTION

A high range resolution profile (HRRP) is the amplitude of coherent summation of the radar returns from target scatterers in each range cell. It contains abundant target information, such as target size, scatterer distribution, etc., and has been widely used for radar automatic target recognition (RATR).

Many kinds of classifiers have been applied to RATR mission and the Bayesian classifier is employed here due to its excellent performance [1, 2]. To use the Bayesian classifier, we should have a good knowledge of the statistical distribution of the HRRP data. So statistical modeling of the HRRP data is the key step in RATR. In our previous work [1, 2], some statistical models were studied and the factor analysis (FA) model obtained the best performance. To smoothly learn the spatial structure of the HRRP data, all our previous models make an assumption that the HRRPs are statistically independent. However, after analyzing the characteristic of radar returns, we find this assumption is inappropriate, since the HRRPs in a sequence are temporally dependent. This temporal dependency can act as target's discriminative information and thus can be used to improve the recognition performance. As in our previous work, here we still concern with the spatial structure of HRRP data based on the Gaussian distribution assumption, but we mainly aim to incorporate the time dependency into modeling. For this purpose, we extend the FA model to the temporal factor analysis (TFA) model [3, 4]. Afterwards, the Variational Bayes method is introduced to perform parameter estimation and model selection for the TFA model [3]. The experiments based on measured data show that the TFA model provides better performance in terms of probability of correct recognition and rejection of confuser target.

The remainder of the text is organized as follows. In Section II we briefly analyze the temporal dependency and the sensitivities of HRRP data. The TFA model and VB method are presented in Section III. Experimental results and

discussion are given in Section IV, with conclusions summarized in Section V.

II. TEMPORAL DEPENDENCY AND SENSITIVITIES OF HRRP DATA

A. Temporal Dependency among HRRPs

Wideband high resolution radars (HRRPs) can spatially resolve a target into a number of scatterers distributed in several range cells. The projection of radar returns from these scatterers onto the radar line of sight (LOS) is interpreted as an HRRP. Formally, an HRRP is defined as:

$$\mathbf{x} = [|x_1|, |x_2|, \dots, |x_d|], \text{ each } x_i = \sum_{j=1}^{L_i} \sigma_{i,j} \exp(j\phi_{i,j}) \quad (1)$$

where d is the number of range cells, L_i is the number of scatterers in the i -th range cell, $\sigma_{i,j}$ and $\phi_{i,j}$ are the reflectivity and phase of the j -th scatterer in the i -th range cell.

Now we will analyze the dependency among the adjacent HRRPs. According to the scattering center model, when there is a little change in target aspect, the reflectivity σ of all scatterers approximately keep invariant whereas the phases ϕ may change greatly. In [1], the range cells are classified into three types. In the first type of range cell, there are lots of small scatterers and no predominant scatterer. All scatterers have very small reflectivity. Since the change of phase is different for each scatterer, the returns from this type of range cell among consecutive HRRPs fluctuate. The second type of range cell consists of a predominant scatterer and a number of small scatterers. The return amplitude of this type of range cell is dependent mainly on the reflectivity of the predominant scatterer and nearly the same for the adjacent HRRPs. In the third type of range cell, there are several predominant scatterers and lots of small scatterers. The return amplitude also depends on the predominant scatterers. It has been proved that the returns from this type of range cell fluctuate significantly [1]. To the radar returns, less fluctuation means more dependency. So the returns from the second type of range cells among adjacent HRRPs are highly dependent. As the observed HRRPs in a sequence are orderly in time and can be viewed as a time series, we refer to the dependency among HRRPs as temporal dependency.

B. Sensitivities of HRRP Data

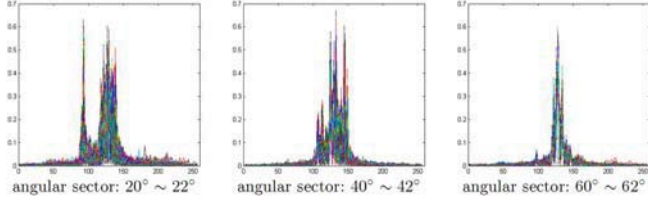


Fig. 1. HRRP sequence of Yak-42 in different angular sectors

Before performing statistical modeling for the HRRP data, the sensitivities of HRRPs must be tackled. They are translation sensitivity, amplitude-scale sensitivity and target-aspect sensitivity.

The HRRP data we used are only a small part intercepted from the radar returns with a range window. The target information is assumed to be completely contained within the HRRPs. Due to the uncertainty of target location, there will be a translation among different HRRPs. This is the translation sensitivity. The HRRPs should be aligned before modeling. Slide correlation alignment is used here for both training and testing data.

Furthermore, the amplitude-scale of an HRRP is a function of target distance and radar parameters such as transmitting power, antenna gain, etc. There may be great difference in amplitude-scale even for the same target when the HRRPs are measured under different conditions. To deal with this amplitude-scale sensitivity, each HRRP data are normalized by dividing its L_2 norm.

The last but not the least is the target-aspect sensitivity. The HRRPs fluctuate dramatically when there are major changes in the target aspect angle (the angle between radar LOS and target orientation). This is shown in Fig. 1 from which we can see the HRRP sequence is nonstationary on the whole. This is disadvantageous for our modeling because the TFA model discussed below can only describe stationary time series. Fortunately, another important information given by Fig. 1 is that the HRRPs change little in a small aspect angular sector. It suggests that the whole HRRP sequence can be divided into several piecewise stationary time series each of which can be modeled by a different TFA model. In this paper, for simplicity, a coarse equal interval partition approach is adopted to segment the whole HRRP sequence into some small pieces which are referred to as frames [1].

Throughout the rest of this paper, the HRRPs in each frame are assumed to have been aligned and normalized..

III. TEMPORAL FACTOR ANALYSIS MODEL

A. Model Description

The temporal factor analysis (TFA) model, also known as linear dynamic model or Kalman filter model, is a widely used statistical model for stationary time series. It is featured by the following state and observation equation:

$$\mathbf{x}_t = \mathbf{A}\mathbf{x}_{t-1} + \mathbf{w}_t, \mathbf{y}_t = \mathbf{C}\mathbf{x}_t + \mathbf{v}_t \quad (2)$$

Time is indexed by discrete t . $\mathbf{y}_t$ denotes the q dimensional

observation vector, i.e. HRRP data here. $\mathbf{x}_t$ denotes the p dimensional hidden state vector. $\mathbf{A}$ is a $p \times p$ transition matrix that describes the temporal evolution of the hidden states. $\mathbf{C}$ is a $q \times p$ observation matrix that relates the state vectors to corresponding observation vectors. Noise in the state $\mathbf{w}_t \sim G(\mathbf{0}, \mathbf{I})$ and noise in the observation $\mathbf{v}_t \sim G(\mathbf{0}, \mathbf{R})$ are assumed uncorrelated random variables where $G(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ denotes Gaussian distribution with mean $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$. $\mathbf{I}$ is identity matrix and $\mathbf{R}$ is a diagonal matrix. The initial state $\mathbf{x}_0$ is also Gaussian distributed with mean $\boldsymbol{\mu}_0$ and covariance matrix $\boldsymbol{\Sigma}_0$. Thus, according to the linear property of Gaussian variable, both $\mathbf{x}_t$ and $\mathbf{y}_t$ follow Gaussian distribution. It should be noted that p is generally much less than q for HRRP data and the model parameters $\{\mathbf{A}, \mathbf{C}, \mathbf{R}, \boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0\}$ are time invariant [2].

The TFA model differs from FA model in the state equation which can be view as a compact representation of the temporal dependency in the high dimensional HRRPs. Let $\mathbf{A}$ equals zero, the TFA model degenerates into the FA model. So it can be regarded as the temporal extension of FA model. Thus, the TFA model can not only describe the spatial characteristic of HRRP data just as FA does, but also capture the temporal dependency present in HRRP sequence.

There is a reason we use TFA model to describe the temporal dependency instead of some other time series models such as autoregressive (AR) model or moving average (MA) model. That is because modeling the high dimensional HRRP sequence is a typical small sample learning problem and a model with too many parameters cannot be estimated accurately from limited HRRP data. Take a first order AR model $\mathbf{y}_t = \mathbf{D}\mathbf{y}_{t-1} + \mathbf{u}_t$ for example, there are $(q+1) \times q$ parameters need to be estimated. However, in a TFA model, we just estimate $(p+1) \times (2p+q)$ parameters which are much less than that in the AR model since p is very small.

B. Model Learning and Classification

In general, all of the model parameters $\{\mathbf{A}, \mathbf{C}, \mathbf{R}, \boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0\}$ are unknown and they have to be estimated from the observed HRRP sequence. When the hidden state dimension p is given, this is a parameter estimation problem and an immediate solution is to resort to maximum likelihood (ML) method. [4] gives a detailed implementation algorithm based on ML for TFA model parameter estimation. However, p is also unknown in advance. How to find the optimal value of p is a model selection problem. Various model selection criteria, such as Akaike information criterion (AIC) and Bayesian information criterion (BIC) have been introduced to solve this problem. In [2], a two phase approach is employed to perform model selection with the help of AIC and BIC. The main drawback

of this approach is its huge computation cost which cannot be afforded when there are many types of targets or the dimension of HRRP is large. To tackle this problem, we adopt a method named Variational Bayes (VB) which is based on Bayesian learning theory and combines automatic model selection with parameter estimation.

In Bayesian learning theory, the central problem is to determine the unknown marginal likelihood $p(\mathbf{y})$ from the observed sample sequence $\mathbf{y} = \{\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_N\}$. The parametric form of $p(\mathbf{y})$ is assumed to be known and denoted as $p(\mathbf{y}|\Theta)$. Θ denotes all parameters in the model and are assumed to be unknown random variables following some prior distribution $p(\Theta)$. $p(\mathbf{y})$ can be obtained by solving the following integral: $p(\mathbf{y}) = \int p(\mathbf{y}|\Theta)p(\Theta)d\Theta$. Unfortunately, this integral is usually an intractable quantity to compute directly and we have to resort to some approximate computation. Variational Bayes is one such approximation method [3]. By applying Jensen's inequality, we can find the lower bound of the log marginal likelihood:

$$\begin{aligned} \ln p(\mathbf{y}) &= \ln \int p(\mathbf{x}, \mathbf{y}, \Theta) d\Theta d\mathbf{x} \\ &= \ln \int q(\mathbf{x}, \Theta) \frac{p(\mathbf{x}, \mathbf{y}, \Theta)}{q(\mathbf{x}, \Theta)} d\Theta d\mathbf{x} \quad (3) \\ &\geq \int q(\mathbf{x}, \Theta) \ln \frac{p(\mathbf{x}, \mathbf{y}, \Theta)}{q(\mathbf{x}, \Theta)} d\Theta d\mathbf{x} \end{aligned}$$

Where $q(\mathbf{x}, \Theta)$ can be any distribution over hidden state and parameters. The VB method assumes that the free distribution can be fully factorized, i.e. $q(\mathbf{x}, \Theta) = q(\mathbf{x})q(\Theta)$. Applying this assumption, (3) can be written as:

$$\begin{aligned} \ln p(\mathbf{y}) &\geq \int q(\mathbf{x})q(\Theta) \ln \frac{p(\mathbf{x}, \mathbf{y}, \Theta)}{q(\mathbf{x})q(\Theta)} d\Theta d\mathbf{x} \\ &= \int \left[\ln \frac{p(\Theta)}{q(\Theta)} + \int q(\mathbf{x}) \ln \frac{p(\mathbf{x}, \mathbf{y}|\Theta)}{q(\mathbf{x})} d\mathbf{x} \right] q(\Theta) d\Theta \\ &= F(q(\Theta), q(\mathbf{x})) \quad (4) \end{aligned}$$

The VB method tries to approach to the true log marginal likelihood through iteratively maximizing this lower bound F with respect to the two free distribution, $q(\mathbf{x})$ and $q(\Theta)$. The implementation algorithm is given as follows:

VBE step:

$$\begin{aligned} q^{(l+1)}(\mathbf{x}) &= \prod_{i=1}^N q^{(l+1)}(\mathbf{x}_i) \quad \text{for } i=1, 2, \dots, N \\ q^{(l+1)}(\mathbf{x}_i) &= \frac{1}{Z_{\mathbf{x}_i}} \exp \left[\int q^{(l)}(\Theta) \ln p(\mathbf{x}_i, \mathbf{y}_i | \Theta) d\Theta \right] \quad (5) \end{aligned}$$

VBM step:

$$q^{(l+1)}(\Theta) = \frac{1}{Z_{\Theta}} p(\Theta) \exp \left[\int q^{(l+1)}(\mathbf{x}) \ln p(\mathbf{x}, \mathbf{y} | \Theta) d\mathbf{x} \right] \quad (6)$$

where superscript l is used to denote iteration number, $\mathbf{y}_i$ and $\mathbf{x}_i$ are the i -th sample in the observed sequence and its corresponding hidden state. $Z_{\mathbf{x}_i}$ and Z_{Θ} are the normalization constant for $q(\mathbf{x}_i)$ and $q(\Theta)$, respectively.

Although arbitrary choices of priors can be made, choosing some specific priors may simplify the computation. For the TFA model, the priors of parameters are defined as follows:

$$\begin{aligned} \mathbf{a}_{(j)} &\sim G(0, \text{diag}(\mathbf{a})^{-1}) \quad \text{for } j=1, 2, \dots, p \\ \mathbf{c}_{(j)} &\sim G(0, \rho_s^{-1} \text{diag}(\boldsymbol{\gamma})^{-1}) \\ \rho_s &\sim Ga(a, b) \quad \text{for } s=1, 2, \dots, q \end{aligned} \quad (7)$$

where $\mathbf{a}_{(j)}$ is the j -th row of $\mathbf{A}$, $\mathbf{c}_{(j)}$ is the j -th column of $\mathbf{C}$, $\mathbf{R}^{-1} = \text{diag}(\boldsymbol{\rho})$, ρ_s is the s -th elements of $\boldsymbol{\rho}$, $Ga(a, b)$ denotes Gamma distribution with shape parameter a and scale parameter b . The form of prior here is motivated by the framework of automatic relevance determination in [5]. p is initialized with its maximum possible value $q-1$. During the VBEM iteration, the elements of redundant rows in $\mathbf{A}$ and columns in $\mathbf{C}$ will approach to zero and only the hidden state dimensions that are required will be retained after learning. This is the automatic model selection process. Readers are referred to [3] for the detailed and exact VBEM algorithm for the TFA model.

After learning the model, the classification task will be carried out. Bayesian classifier is employed here. For a c -class classification task, the testing HRRP sequence $\mathbf{y} = \{\mathbf{y}_1, \mathbf{y}_2, \dots, \mathbf{y}_N\}$ will be classified to the k -th class with the maximum a posterior, i.e. $k = \arg \max_j p(j|\mathbf{y})$, $j=1, 2, \dots, c$. According to the Bayes' theorem, $p(j|\mathbf{y})$ is proportional to $p(\mathbf{y}|j)p(j)$. We assume the prior $p(j)$ is the same for all the c classes, so the central problem here lies in computing the conditional likelihood of the testing HRRP sequence. Given a TFA model with parameters $\{\mathbf{A}, \mathbf{C}, \mathbf{R}, \boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0\}$, the log conditional likelihood is :

$$\log p(\mathbf{y}|j) = -\frac{1}{2} \sum_{t=1}^N \left\{ \log |\Sigma_{e_t}| + \mathbf{e}_t^T \Sigma_{e_t}^{-1} \mathbf{e}_t \right\} - \frac{Np}{2} \log(2\pi) \quad (8)$$

where prediction error $\mathbf{e}_t$ and corresponding covariance matrix Σ_{e_t} are computed through the Kalman filter recursions given in [4]. By contrast, the log likelihood in the FA model is the same as (8) except for a different definition of Σ_{e_t} and $\mathbf{e}_t : \Sigma_{e_t} = \mathbf{C}\mathbf{C}^T + \mathbf{R}$, $\mathbf{e}_t = \mathbf{y}_t - \boldsymbol{\mu}_{ij}$, where $\boldsymbol{\mu}_{ij}$ is the mean of the training HRRPs in the i -th frame of the j -th target.

IV. EXPERIMENTAL RESULTS AND DISCUSSION

A. Data Overview

The results presented in this paper are based on measured data. The center frequency and bandwidth of the radar are 5520MHz and 400MHz, respectively. The projections of target trajectories onto ground are shown in Fig. 2. As it is seen from Fig. 2, the measured data are divided into several parts that are labeled by numbers. In the experiments, the fifth and sixth parts of An-26, the sixth and seventh parts of Cessna, the second and fifth parts of Yak-42 are chosen as training data while the other parts are taken as testing data. The training

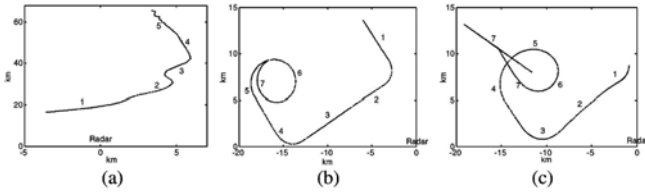


Fig. 2. Projections of target trajectories onto ground. (a) Yak-42; (b) Cessna; (c) An-26.

TABLE I. CONFUSION MATRICES

	FA			TFA		
	An-26	Cessna	Yak-42	An-26	Cessna	Yak-42
An-26	93.16%	4.83%	2.01%	95.7%	2.34%	1.96%
Cessna	0.65%	97.54%	1.81%	1.19%	98.79%	0.2%
Yak-42	1.09%	0.51%	98.40%	0.71%	0.28%	99.1%
average	96.08%			97.84%		

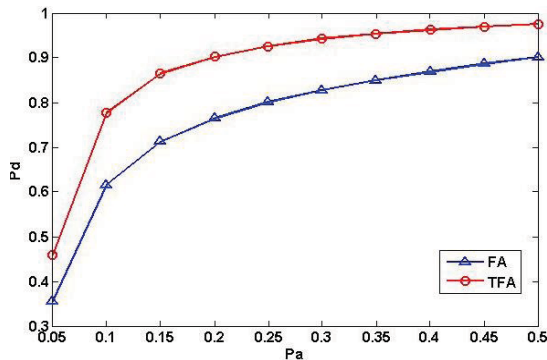


Fig. 3. ROC curves for the FA model and the TFA model.

data are divided into equal length frames, each containing 1000 HRRPs. Thus, there are 52 frames for An-26, 52 frames for Cessna and 44 frames for Yak-42.

B. Recognition Performance

In table 2 we present the confusion matrices and average correct recognition rates of the FA and the TFA model for comparison. The length of each testing HRRP sequence is 3. We can see from the table that the average correct recognition rate of the TFA model is approximately 1.8 percentage points larger than that of the FA model

C. Rejection Performance

Besides correct recognition rate, the ability to reject unseen confuser targets is also an important indicator of model performance. So we also perform a rejection experiment. 18000 truck HRRPs generated by the simulation software XPATCH are used as confuser target. The receiver operating characteristic (ROC) curves of the two models are plotted in Fig.3. P_a denotes the probability of false alarm while P_d is the corresponding correct detection probability. Fig. 3 illuminates that the correct detection probability of the TFA model is higher for a given P_a .

D. Discussion

Different targets may have similar HRRPs, so there is misclassification. But the temporal characteristic between adjacent HRRPs is likely to be more target dependent. That is the reason why TFA model has the better performance. The experiments verify our statement that the HRRPs in a sequence are temporal correlated rather than independent. The TFA model which utilizes both spatial and temporal characteristic of HRRP data is more suitable for HRRP modeling than the FA model.

V. CONCLUSION

In this paper, we propose a temporal factor analysis model which furthers previously studied factor analysis model to incorporate the temporal dependency among HRRPs for HRRP modeling. Variational Bayes method has been introduced to learn the model. The results show our proposed model outperforms the FA model both in recognition and rejection performance.

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